Physical-Layer Demodulation Benchmark

Bit Error Rate (BER) & Cryptographic Authorization Report

CONFIDENTIAL
SUBJECT TO MNDA

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Author: John Murphy

1. Executive Summary & Verification Metrics

This technical evaluation report validates the physical-layer communication reliability, frame synchronization robustness, and Bit Error Rate (BER) performance of the Quackworks Acoustic Lock (QAL) ultrasonic physical layer.

An automated Software-in-the-Loop (SIL) verification testbench executed 50 consecutive cryptographic token transmissions. Each burst was modulated into 16.0 kHz / 17.0 kHz continuous-phase audio, processed through the bandpass filtering and quadrature demodulation pipeline, verified against IEEE 802.3 CRC-32 integrity polynomials, and authenticated by the simulated automotive Electronic Control Unit (FakeECU) root vault.

PHYSICAL FRAME ACCURACY	BIT ERROR RATE (BER)	ECU PROPULSION GATING
100.0% Pass	**0.00% BER**	**100.0% Pass**
50 / 50 Frame Locks	0 / 19,600 Bits Corrupted	50 / 50 Stage 2 Authorized

2. Physical-Layer Modulation Spec

- **Modulation Scheme:** Continuous-Phase FSK (CP-FSK)
- **Sampling Frequency:** 48,000 Hz (16-bit PCM)
- **Carrier Space (f0):** 16,000 Hz (Binary '0')
- **Carrier Mark (f1):** 17,000 Hz (Binary '1')
- **Symbol Duration:** 10.0 ms (100 Baud rate)
- **Preamble:** 32-bit alternating clock synchronization
- **Start of Frame (SOF):** 16-bit Barker-like code (0xE38E)
- **Demodulator:** Dual-carrier orthogonal quadrature correlator (7.0 ms integration + 1.5 ms echo guard)

3. Cryptographic Framing Spec

- **Frame Size:** 49 bytes (392 payload bits)
- **Slot Identifier:** 1 Byte (0x01 Master Owner)
- **Ephemeral Nonce:** 8 Bytes (64 bits, CSPRNG minted)
- **Temporal Epoch:** 4 Bytes (UTC Unix timestamp window)
- **Signature:** 32 Bytes HMAC-SHA256 (HSM derived key)
- **Frame Integrity:** 4 Bytes CRC-32 (IEEE 802.3 standard)
- **Replay Defense:** Consumed nonce tracking registry
- **Functional Safety:** ASIL standstill interlock (v <= 0.1 km/h)

4. Security & Functional Safety Verification

1. **Replay Protection:** All 50 nonces were verified as cryptographically unique and marked as consumed within the non-volatile registry. Any re-injection of previous frames produced DTC 0x06 (Nonce Replay).
2. **Forensic Black Box Audit:** Every authorization was sealed into the SHA-256 hash-chained diagnostic ledger, linking the timestamp, GPS coordinates, method (Acoustic CPFSK), and status.
3. **Functional Safety (ISO 26262):** Standstill invariants were verified ($v \leq 0.1$ km/h) prior to Stage 2 authorized contactor gating. Fail-silent dormancy is enforced while rolling so driving torque is never interrupted.

Engineering Sign-off:
John Murphy
Founder & Lead Inventor | Quackworks Audio IP
Direct Contact: Mail@quackworkstudios.com

Compliance Status:
ISO/SAE 21434 & UNECE WP.29 R155 Aligned
Bench Verification: Verified 0.00% BER
Portal: quackworkstudios.com/qal

5. Empirical 50-Burst Validation Log Table

The following log records the sequential evaluation executed on 24 September 2026. Every burst used an independently generated cryptographic nonce, confirming both physical demodulation accuracy (0.00% BER) and anti-replay verification:

Burst	Timestamp (UTC)	Nonce (Hex)	CRC-32	Bits	Demod	ECU State
01	2026-09-24 13:12:19	AFB6537F…	0x17D38A30	392	PASS	AUTHORIZED
02	2026-09-24 13:12:19	D0842F7D…	0x17335A97	392	PASS	AUTHORIZED
03	2026-09-24 13:12:19	7F8FF18E…	0xE1E01203	392	PASS	AUTHORIZED
04	2026-09-24 13:12:19	25D24024…	0xECB31FDC	392	PASS	AUTHORIZED
05	2026-09-24 13:12:19	3987A2FF…	0x739EDDFB	392	PASS	AUTHORIZED
06	2026-09-24 13:12:19	62FE24A8…	0x1EAD3C61	392	PASS	AUTHORIZED
07	2026-09-24 13:12:19	F633BC15…	0x9BD423ED	392	PASS	AUTHORIZED
08	2026-09-24 13:12:19	052F8E73…	0x4DB738E1	392	PASS	AUTHORIZED
09	2026-09-24 13:12:19	734DA8D0…	0xD0F685BA	392	PASS	AUTHORIZED
10	2026-09-24 13:12:19	DFBB8D1F…	0x64102352	392	PASS	AUTHORIZED
11	2026-09-24 13:12:20	DDC58215…	0xFDE0503A	392	PASS	AUTHORIZED
12	2026-09-24 13:12:20	BF9BA286…	0x7EA5BA78	392	PASS	AUTHORIZED
13	2026-09-24 13:12:20	55DEAE89…	0xA37FBBCC	392	PASS	AUTHORIZED
14	2026-09-24 13:12:20	522B9857…	0xDE7B7CFA	392	PASS	AUTHORIZED
15	2026-09-24 13:12:20	F53096DF…	0x00C1C5F0	392	PASS	AUTHORIZED
16	2026-09-24 13:12:20	833BDAB0…	0xBA886B95	392	PASS	AUTHORIZED
17	2026-09-24 13:12:20	6461DB0A…	0x4E13EE6D	392	PASS	AUTHORIZED
18	2026-09-24 13:12:20	34869A46…	0x0E6F3A0D	392	PASS	AUTHORIZED
19	2026-09-24 13:12:20	56B4FA5E…	0xBCEEE8F2	392	PASS	AUTHORIZED
20	2026-09-24 13:12:20	168B591B…	0xF96BA773	392	PASS	AUTHORIZED
21	2026-09-24 13:12:20	31EA7142…	0x3CCB2AC0	392	PASS	AUTHORIZED
22	2026-09-24 13:12:20	11CA8B3F…	0x966452AF	392	PASS	AUTHORIZED
23	2026-09-24 13:12:20	F74CF543…	0x2FF7C1D9	392	PASS	AUTHORIZED
24	2026-09-24 13:12:20	5AA224B4…	0xEFFD5B91	392	PASS	AUTHORIZED
25	2026-09-24 13:12:20	4BC0FC2F…	0x2CF6453C	392	PASS	AUTHORIZED
26	2026-09-24 13:12:20	7944DEA8…	0xCC7EC9B9	392	PASS	AUTHORIZED
27	2026-09-24 13:12:20	C363D83C…	0xF81ED057	392	PASS	AUTHORIZED
28	2026-09-24 13:12:20	5C0FAB11…	0x33E23909	392	PASS	AUTHORIZED
29	2026-09-24 13:12:20	64DAA7E7…	0xCCDBFD4D	392	PASS	AUTHORIZED
30	2026-09-24 13:12:20	600A2672…	0xBA897E39	392	PASS	AUTHORIZED
31	2026-09-24 13:12:20	0736C53E…	0xB8B6B03A	392	PASS	AUTHORIZED
32	2026-09-24 13:12:20	C6B0D7C1…	0x838531A6	392	PASS	AUTHORIZED
33	2026-09-24 13:12:20	A791D2BB…	0xB6EBE4C0	392	PASS	AUTHORIZED
34	2026-09-24 13:12:20	C3A9F596…	0x99C68DA9	392	PASS	AUTHORIZED
35	2026-09-24 13:12:20	448F1307…	0x11BDD28A	392	PASS	AUTHORIZED
36	2026-09-24 13:12:20	0454CE0C…	0x11469BA6	392	PASS	AUTHORIZED
37	2026-09-24 13:12:20	D103BA84…	0x243015FC	392	PASS	AUTHORIZED
38	2026-09-24 13:12:20	20280F59…	0x26FC092E	392	PASS	AUTHORIZED
39	2026-09-24 13:12:20	A6EAA45B…	0xA86F241A	392	PASS	AUTHORIZED
40	2026-09-24 13:12:20	A618A7CA…	0xC468EA4B	392	PASS	AUTHORIZED
41	2026-09-24 13:12:20	915CF466…	0x5E47AD58	392	PASS	AUTHORIZED
42	2026-09-24 13:12:20	E2BBB050…	0xEAB3AA4D	392	PASS	AUTHORIZED
43	2026-09-24 13:12:20	AE2FE911…	0x09AD4379	392	PASS	AUTHORIZED
44	2026-09-24 13:12:20	654EF1D2…	0x6ED0BA2F	392	PASS	AUTHORIZED
45	2026-09-24 13:12:20	4370DE98…	0xB4E9DBCE	392	PASS	AUTHORIZED
46	2026-09-24 13:12:20	A6389839…	0x300461D1	392	PASS	AUTHORIZED
47	2026-09-24 13:12:20	206B7700…	0x2E5007D1	392	PASS	AUTHORIZED
48	2026-09-24 13:12:20	94B483C8…	0x3E021196	392	PASS	AUTHORIZED
49	2026-09-24 13:12:20	98FF8D8D…	0x2944AD24	392	PASS	AUTHORIZED
50	2026-09-24 13:12:21	80F8D453…	0xEF61504C	392	PASS	AUTHORIZED

Total Bursts: 50 / 50 **Bit Errors:** 0 / 19,600 **FSR:** 100.0% Pass **Final BER:** **0.00%**